

Newton's Method

Finding Roots Using Tangent Lines

Learning Objectives

By the end of this lecture, students will be able to:

- Explain the tangent-line idea behind Newton's Method
- Derive the iteration formula from the tangent line
- Use Newton's Method to approximate a root
- Name common situations when the method can fail

Why Root Finding?

Some equations cannot be solved exactly with algebra.

For example:

$$x^2 - 2 = 0$$

The exact solution is:

$$x = \sqrt{2}$$

Newton's Method gives a numerical way to approximate this solution.

Geometric Idea

Start with an initial guess:

$$x_0$$

At each guess, draw the tangent line to the curve:

$$y = f(x)$$

The tangent line crosses the x-axis at a new point, which becomes:

$$x_{n+1}$$

Repeat this process to improve the estimate.

Tangent Line at x_n

The tangent line to $y = f(x)$ at the point $(x_n, f(x_n))$ is:

$$y = f(x_n) + f'(x_n) (x - x_n)$$

Here:

- x_n is the current estimate
- $f(x_n)$ is the function value at x_n
- $f'(x_n)$ is the slope at x_n

Finding the Next Guess

To find the next estimate x_{n+1} , set the tangent line equal to zero.

That gives:

$$0 = f(x_n) + f'(x_n) (x_{n+1} - x_n)$$

Solve this equation for x_{n+1} .

Newton's Iteration Formula

Solving the equation gives:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This means:

- start with x_n
- compute $f(x_n)$
- compute $f'(x_n)$
- use the formula to get x_{n+1}

Example: $f(x) = x^2 - 2$

For this function:

$$f(x) = x^2 - 2$$

and:

$$f'(x) = 2x$$

Newton's update becomes:

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n}$$

Start with:

$$x_0 = 1$$

Iteration Results

Using the formula with $x_0 = 1$:

$$x_1 = 1 - \frac{1^2 - 2}{2 \cdot 1} = 1.5$$

$$x_2 = 1.5 - \frac{1.5^2 - 2}{2 \cdot 1.5} \approx 1.4167$$

$$x_3 = 1.4167 - \frac{1.4167^2 - 2}{2 \cdot 1.4167} \approx 1.4142$$

The estimates quickly approach:

$$\sqrt{2} \approx 1.4142$$

When Newton's Method Works Well

Newton's Method works well when:

- the starting guess x_0 is close to the root
- the derivative $f'(x_n)$ is not zero
- the function behaves smoothly near the root

In these cases, each step usually gives a better estimate.

When Newton's Method Can Fail

Possible problems include:

- a poor initial guess that is far from the root
- a horizontal tangent where $f'(x_n) = 0$
- the next guess jumping far from the root
- the method entering a cycle and repeating values

A good starting point makes the method more reliable.

Interactive Animation Idea

An animation can show how each step works:

- plot the curve

$$f(x) = x^2 - 2$$

- mark the current estimate x_n
- draw the tangent line at $(x_n, f(x_n))$
- show the x-intercept as x_{n+1}

Students can step through the iterations one by one.

Summary

- Newton's Method approximates roots of

$$f(x) = 0$$

- It uses tangent lines to move from x_n to x_{n+1}
- The formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- It is best when the guess is good and the function is smooth.